

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic

Data Science Baseline Modelling Report

By

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Track: Data Science

Prepared for the AnalystLab Africa Experience Lab

Week 5 Submission – Data Preparation, Feature Engineering & Baseline Model Development

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1. Introduction

This report documents the Data Science track's data preparation, feature engineering and baseline modelling work on the HealthConnect no-show prediction problem, carried out in the notebook `wk5_data_prep_feature_engr_baseline_model_dev_REVISED.ipynb`. It builds directly on the target variable, candidate feature set and modelling approach proposed in the Week 4 Machine Learning Problem Definition Document, and follows the Week 4 brief's staged project journey from Problem Understanding into Analysis and Solution Design.

This version of the notebook extends the work previously reported in two important ways: it adds a formal battery of independent-samples t-tests to statistically validate the relationships suggested by the earlier correlation analysis, and it documents an actual cross-track exchange with the Data Analytics track, whose dashboard findings independently corroborate the Data Science track's own results. Both are incorporated into this report, in Sections 8 and 9 respectively.

A documentation issue identified in the previous version of this report, a mismatch between the notebook's own narrative commentary and its executed code, has been **partially corrected** in this revision: the reported baseline metrics and confusion matrix now match the executed output throughout the notebook. However, the notebook's auto-generated README still describes the train/test split as a stratified `train_test_split`, when the executed code demonstrably uses a patient-aware `GroupShuffleSplit`. This remaining inconsistency is addressed in Section 9.

2. Week 4 Problem Review

The Week 4 Machine Learning Problem Definition Document framed the HealthConnect challenge as a supervised binary classification problem: predicting, at the time of booking, whether a scheduled appointment will result in a no-show. Three design decisions from Week 4 were carried into this stage of work and are confirmed, unchanged, by the notebook:

- **Target definition:** the binary target `no_show` (1 = No-Show, 0 = Attended) is derived from `appointment_outcome`, with Cancelled appointments excluded from the primary modelling population, consistent with the Week 4 rationale that a cancellation is a distinct, proactive patient action.
- **Leakage exclusion of `waiting_time_minutes`:** the Week 4 document provisionally excluded this field pending clarification of its meaning. The Week 5 notebook resolves this conclusively: `waiting_time_minutes` is confirmed as information that only becomes available during or after the appointment, and is therefore excluded from the feature set as temporal data leakage.
- **Candidate features:** the strongest Week 4 candidates, `previous_no_shows` and `booking_lead_days`, are confirmed in Week 5 as the two most informative predictors identified so far (see Section 8).

Two aspects of the Week 5 work go beyond what Week 4 specified and represent a genuine methodological improvement: a patient-aware train/test split (Section 5), which was not explicitly proposed in Week 4, and a newly identified data-quality anomaly involving Sunday appointments (Section 3), neither of which had been surfaced at the Week 4 problem-definition stage.

3. Data Preparation

The HealthConnect Appointment Dataset (5,000 rows, 18 columns) was loaded and inspected as a working copy (week5). Data preparation addressed data types, missing values, duplicate and consistency checks, and one previously undocumented anomaly.

3.1. Data Types

booking_date and appointment_date were converted from object (text) to datetime, enabling date-based feature engineering and consistency checks. All other columns retained their original, appropriate types.

3.2. Missing Values

Column	Missing	% Missing	Treatment
reminder_channel	1,366	27.32%	Imputed as 'No Reminder' – structural missingness, confirmed by cross-tabulation against reminder_sent (all 1,366 missing rows had reminder_sent = 'No')
distance_to_clinic_km	90	1.80%	Imputed with the training-safe median (8.7 km) – information known at booking time, so imputation does not introduce leakage
waiting_time_minutes	60	1.20%	Not imputed – excluded entirely from the feature set as a data-leakage risk (see Section 3.4)

3.3. Duplicates and Identifier Checks

No fully duplicated rows were found. appointment_id was confirmed unique across all 5,000 rows (a valid primary key), while patient_id had 1,696 unique values, confirming that most patients have more than one recorded appointment.

3.4. Consistency Checks and Leakage Assessment

Three internal consistency checks were run and all passed cleanly: no record had previous_no_shows greater than previous_appointments; no record had a booking_date after its appointment_date; and the supplied booking_lead_days matched the calculated difference between the two dates for all 5,000 rows exactly (0 mismatches).

A structured leakage assessment was then documented, consistent with recommended practice for avoiding target and temporal leakage in predictive modelling (Kaufman et al., 2012). Four categories of field were excluded from the predictor set: *appointment_outcome* (the source of the target itself), *waiting_time_minutes* (only known during or after the appointment), *appointment_id* (a record identifier with no generalisable meaning), and *patient_id* (retained only as a grouping variable for the train/test split, not as an ordinary predictor).

3.5. Sunday Appointments Anomaly

A previously undocumented data-quality issue was identified during preparation: 737 appointments (14.7% of the dataset) are recorded on a Sunday, despite the HealthConnect Clinic Knowledge Base stating that the clinic is closed on Sundays and public holidays. Because the dataset is fictional and the discrepancy could not be resolved against an authoritative source, the records were retained rather than removed, and a binary is_sunday_appointment flag was created so the model can learn from, rather than silently absorb, this anomaly. This finding should be escalated to the data owner, consistent with risk R2 in the Week 4 Risk and Dependency Register.

4. Feature Engineering

Beyond the variables already present in the dataset, two categories of new feature were engineered directly from existing fields, both created only from information available at, or before, the time of booking.

- **is_sunday_appointment (binary):** flags whether an appointment falls on a Sunday, allowing the model to learn any distinct behavioural pattern associated with the anomaly described in Section 3.5, rather than treating these records as ordinary weekday appointments.
- **Date-derived features (numerical, from booking_date and appointment_date):** booking_month, booking_day_of_week, booking_day_of_year, booking_week_of_year, appointment_month, appointment_day_of_year and appointment_week_of_year. These capture seasonal and weekly booking patterns (for example, whether bookings made in a particular month or week are more prone to no-shows) that are not visible from booking_lead_days alone.

The redundant calculated_lead_days column, created only to validate booking_lead_days in Section 3.4, was dropped before feature engineering began, since it duplicated an existing field exactly (0 mismatches) and carried no additional information.

All engineered features were created before the train/test split described in Section 5, but critically, they were derived only from booking_date and appointment_date, both of which are known in full at the moment a prediction would be required. No engineered feature introduces information that would be unavailable at booking time.

5. Train/Test Strategy

Because the dataset contains repeated appointments for the same 1,696 patients, a conventional row-level random split risks placing different appointments from the same patient into both the training and test sets, which would let the model implicitly learn patient-specific patterns it should not have access to at test time. This is a well-documented risk in clinical machine learning, where record-level cross-validation can substantially overstate real-world performance compared with a subject-wise (patient-aware) split (Saeb et al., 2017).

A patient-aware split was therefore used: scikit-learn's GroupShuffleSplit, grouped by patient_id, with an 80/20 train/test proportion and a fixed random seed for reproducibility (Pedregosa et al., 2011). This produced 3,771 training records and 966 test records, and an explicit overlap check confirmed 0 patients appearing in both sets. The resulting class balance was close to even in both partitions (51.4% no-show in training, 50.0% in test), so no additional resampling was applied at this stage.

Preprocessing was fitted only after the split, to avoid preprocessing leakage: categorical variables were one-hot encoded (with the first category of each dropped to avoid redundant columns) separately on the training and test sets, with test columns then aligned to the training column structure; and StandardScaler was fitted only on the training data, with the same fitted parameters subsequently applied to transform the test data. The final encoded feature matrix contained 35 columns.

6. Baseline Model

Logistic regression was selected as the baseline model (scikit-learn's LogisticRegression with the liblinear solver; Pedregosa et al., 2011), for four reasons consistent with standard baselining practice for binary classification problems (James et al., 2021):

- **Simplicity and interpretability:** as a linear model, its coefficients directly indicate the direction and relative strength of each feature's association with no-show risk.
- **Computational efficiency:** it trains quickly, making it a practical first benchmark before investing in more complex models.

- **Native fit for binary classification:** it is designed for exactly this kind of two-class problem and outputs calibrated-looking probabilities, not just hard class labels, which matters for the risk-score output specified in the Week 4 ML System Architecture Design.
- **A meaningful comparison point:** it establishes a benchmark against which the more advanced models proposed for Week 6 (Section 10) can be fairly judged.

The model was trained on the 3,771-record training set (35 encoded and scaled features) and used to generate both class predictions and no-show probabilities on the 966-record held-out test set.

7. Initial Evaluation

The figures below are taken directly from the notebook's executed evaluation cell and are the figures this report treats as authoritative (see the note in Section 9 on a documentation discrepancy found elsewhere in the notebook).

Metric	Value
Accuracy	0.6366 (63.66%)
Precision (No-Show)	0.6304 (63.04%)
Recall (No-Show)	0.6605 (66.05%)
F1-score (No-Show)	0.6451 (64.51%)
ROC-AUC	0.6873 (68.73%)

The confusion matrix on the 966-record test set (483 Attended, 483 No-Show) was:

	Predicted Attended	Predicted No-Show
Actual Attended	296 (True Negative)	187 (False Positive)
Actual No-Show	164 (False Negative)	319 (True Positive)

8. Interpretation

An accuracy of 63.66% and ROC-AUC of 0.6873 place the baseline clearly above random guessing (50% accuracy, 0.50 ROC-AUC) but well short of a strong classifier. In practical terms, the model correctly classifies roughly 64 appointments in every 100, and can moderately, but not strongly, separate patients likely to no-show from those likely to attend.

The precision/recall balance is informative for how the clinic might use this model: recall (66.05%) is somewhat higher than precision (63.04%), meaning the model catches about two-thirds of genuine no-shows, at the cost of a meaningful number of false alarms (187 patients flagged as likely no-shows who would, in fact, have attended). Given that the Week 4 architecture proposes using risk scores to prioritise outreach such as a confirmation call, rather than to cancel or reassign appointments automatically, this recall-leaning balance is a reasonable starting point: the operational cost of an unnecessary reminder call is low relative to the cost of an unanticipated empty slot.

8.1. Statistical Validation of the Candidate Predictors

The correlation analysis carried out in Week 5 was followed up with five independent-samples t-tests, formally comparing the mean value of each numerical candidate feature between the Attended and No-Show groups at a significance level of $\alpha = 0.05$. The results are summarised below.

Variable	Mean (Attended)	Mean (No-Show)	p-value	Significant?
previous_no_shows	0.46	0.64	1.62×10^{-17}	Yes
booking_lead_days	24.52	34.53	1.07×10^{-90}	Yes
distance_to_clinic_km	9.65 km	10.50 km	8.94×10^{-6}	Yes
previous_appointments	2.95	3.10	3.48×10^{-3}	Yes
age	49.25	48.22	0.050	No (borderline)
waiting_time_minutes (informational only)	24.29 min	24.20 min	0.781	No

These results give the earlier correlation evidence formal statistical backing rather than leaving it as a descriptive observation. `previous_no_shows` and `booking_lead_days` are confirmed as highly significant, consistent with the wider no-show literature identifying a patient's own no-show history and booking lead time as the most reliable predictors of future non-attendance (Dantas et al., 2018). `distance_to_clinic_km` and `previous_appointments` are also statistically significant, though with smaller mean differences that suggest a more modest practical effect. `age` sits essentially on the conventional 0.05 threshold ($p = 0.050$) and is treated as not reliably significant, consistent with its weak correlation ($r = -0.05$) reported previously. The `waiting_time_minutes` test, run for information only, found no significant difference between groups ($p = 0.781$), which is consistent with, and adds confidence to, the decision to exclude it from the model on data-leakage grounds (Section 3.4) rather than on the basis of it being predictive.

8.2. Cross-Track Validation with Data Analytics

The notebook also documents a direct exchange with the Data Analytics track, whose dashboard-based analysis of the same appointment population independently reported: no-show rates rising from 29.5% for appointments booked 0–7 days ahead to 63.9% for those booked 31–60 days ahead; a no-show rate of 57.8% associated with prior no-show history; 60.5% associated with longer travel distances; and 54.6% associated with the absence of a reminder. These figures, produced independently using a different tool and framing (rates rather than correlation coefficients or t-tests), corroborate the Data

Science track's own findings on `booking_lead_days`, `previous_no_shows` and `distance_to_clinic_km`, and represent a genuine, documented instance of cross-track validation rather than a single track's analysis standing alone.

9. Modelling Limitations

- **Documentation inconsistency, partially resolved:** the baseline metrics reported in the notebook's markdown commentary and auto-generated README now match the executed code's output throughout (Section 7), correcting the discrepancy identified in the previous version of this report. However, the README's description of the train/test methodology has not been updated to match: it still states that a stratified `train_test_split` (80/20, `random_state=42`) was used, when the executed code demonstrably uses a patient-aware `GroupShuffleSplit` grouped by `patient_id` (Section 5). This should be corrected before the notebook or its README is shared further.
- **Categorical variables not yet formally tested:** the Week 5 statistical work (Section 8.1) covers only numerical variables via t-tests. Categorical variables such as `gender`, `appointment_type` and `reminder_sent` have not yet been tested for association with no-show status using an appropriate method such as a chi-square test of independence.
- **Baseline ceiling:** as a linear model, logistic regression cannot capture non-linear or threshold-based relationships that the exploratory boxplots hinted at, particularly for `previous_no_shows` and `distance_to_clinic_km`.
- **Precision/recall trade-off not yet tuned:** the default 0.5 classification threshold was used throughout; no work has yet been done to calibrate this threshold against the clinic's actual cost of a false positive (an unnecessary reminder) versus a false negative (a missed intervention opportunity).
- **Exclusion of waiting_time_minutes:** correctly excluded to prevent leakage, and the Week 5 t-test (Section 8.1) found no significant difference in waiting time between the two groups in any case, so its exclusion is unlikely to have discarded a strong signal, though this does not rule out its relevance to a differently framed prediction task (for example, predicting cancellation after a short in-clinic wait).
- **Sparse gender category:** `gender` contains three categories (Female, Male and a small 'Other' category); the 'Other' category is sparse enough that its one-hot encoded column may carry very little statistical weight and should be monitored in Week 6 rather than assumed to be reliably estimated.
- **Unresolved Sunday-appointment anomaly:** `is_sunday_appointment` was engineered to let the model learn from the pattern, but the underlying cause of the anomaly (data error versus an undocumented clinic exception) remains unconfirmed and should be raised with the data owner.
- **Synthetic dataset:** as with the Week 4 assessment, findings should be treated as illustrative of methodology on this fictional Experience Lab dataset rather than as a validated clinical model.

10. Recommendations for Week 6

1. **Trial non-linear candidate models:** evaluate Random Forest and gradient boosting approaches (e.g., XGBoost, LightGBM) against the same patient-aware split, to test whether they can extract more value from the weakly linear features identified in Section 8.
2. **Extend statistical testing to categorical variables:** apply chi-square tests of independence to `gender`, `appointment_type`, `reminder_sent` and other categorical fields, to complete the statistical validation started with the Week 5 t-tests (Section 8.1).

3. **Use group-aware cross-validation for tuning:** hyperparameter tuning should use GroupKFold (grouped by patient_id), not standard K-Fold, so that the leakage protection built into the train/test split in Week 5 is preserved throughout model selection, not just at the final evaluation step (Saeb et al., 2017).
4. **Calibrate the decision threshold:** work with clinic stakeholders to establish the relative cost of a false positive versus a false negative, and select an operating threshold (rather than the default 0.5) accordingly; consider a precision-recall curve alongside the ROC curve for this purpose.
5. **Finish correcting the notebook documentation:** update the README's description of the train/test split to reflect the patient-aware GroupShuffleSplit actually used, now that the reported metrics themselves have been corrected (Section 9).
6. **Continue the Data Analytics collaboration:** use the dashboard-derived rates shared this week (Section 8.2) as an ongoing cross-check as modelling progresses, and share Week 6 model outputs back with that track so both sets of findings continue to be jointly validated.
7. **Escalate the Sunday-appointment anomaly:** raise the 737 Sunday appointments with the data owner or mentor to confirm whether this reflects a genuine clinic exception or a data-quality error.
8. **Monitor sparse categories:** check the encoded 'Other' gender category and any other sparse one-hot columns for stability across cross-validation folds before relying on their coefficients or importances.

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